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BATCH - 3 [DS]

ASSIGNMENT - 1

Question - 1

A scalar LSTM ..... $x_t = 0.5$, with hidden state $h_{t-1} = 0.2$ ..... Compute, f_t, i_t, $\tilde{c}_t$, c_t, o_t & i_t.

Ans $x_t = 0.5$ $\quad h_{t-1} = 0.2$ $\quad c_{t-1} = 0.4$

⊙ $f_t = \sigma[W_f \cdot x_t + U_f \cdot h_{t-1} + b_f)$

$= \sigma((0.7 \times 0.5) + (0.3 \times 0.2) + 0.1)$

$= \sigma(0.5100)$

$\therefore \sigma(0.5100) = \frac{1}{1 + e^{(-0.5100)}}$

$= \frac{1}{1.6005} = \underline{0.6248}$

$\therefore \boxed{f_t = 0.6248}$

⊙ $i_t = \sigma(W_i \cdot x_t + U_i \cdot h_{t-1} + b_i)$

$= \sigma((0.6 \times 0.5) + (0.4 \times 0.2) + 0.0)$

$= \sigma(0.3800)$

$\Rightarrow \sigma(0.3800) = \frac{1}{1 + e^{-(0.3800)}} = \frac{1}{1.6839}$

$= 0.5939$

$\therefore \boxed{i_t = 0.5939}$

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⊙ $\tilde{C}_t = \tanh(W_c \cdot x_t + U_c \cdot h_{t-1} + b_c)$

$\tanh((0.9 \times 0.5) + (0.2 \times 0.2) + 0.0)$

$\tanh(0.4900)$

$$\tanh(0.4900) = \frac{e^{z} - e^{-z}}{e^{z} + e^{-z}}$$

$$= \frac{e^{(0.4900)} - e^{(-0.4900)}}{e^{(0.4900)} + e^{(-0.4900)}} = 0.4542$$

$\therefore \boxed{\tilde{C}_t = 0.4542}$

⊙ $O_t = \sigma(W_o \cdot x_t + U_o \cdot h_{t-1} + b_o)$

$\sigma((0.5 \times 0.5) + (0.5 \times 0.2) + 0.1)$

$\sigma(0.4500)$

$$\sigma(0.4500) = \frac{1}{1 + e^{-(0.4500)}} = 0.6106$$

$\therefore \boxed{O_t = 0.6106}$

⊙ $C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t$

$= (0.6248 \times 0.4) + (0.5939 \times 0.4542)$

$\boxed{C_t = 0.5197}$

⊙ $h_t = \tanh(C_t) \odot O_t$

$\tanh(0.5197) \times 0.6106$

$h_t = 0.4775 \times 0.6106$

$\boxed{h_t = 0.2916}$

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Question-2

(a) Using the standard ....., how many trainable parameters does the layer have?

Ans Input Dimension (x_{dim}) = 10
Hidden units (h_{dim}) = 32

For a single gate, linear trasformation is:

$$gate = \sigma(Wx_t + Uh_{t-1} + b)$$

⊙ Input Weight: Parameters = $h_{dim} \times x_{dim}$
$= 32 \times 10 = 320$

⊙ Recurrent/Hidden Weight: Parameters = $h_{dim} \times h_{dim}$
$= 32 \times 32$
$= 1024$

⊙ Bias: One bias per hidden unit
Parameters = $h_{dim} = 32$

Total Trainable Parameter per gate = $320 + 1024 + 32$
$= 1376$

∵ We have 4 gates
∴ Total Trainable Parameters = $4 \times 1376 = 5504$

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(b) Framework ...., How many parameter does the layer have under that convention.

Ans In such frameworks, the transformation for each gate includes two independent Bias - one for input projection & one for hidden state projection.

- Input Weight = $32 \times 10 = 320$
- Hidden Weight = $32 \times 32 = 1024$
- Input Bias = 32
- Hidden Bias = 32

Total parameters per gate = $320 + 1024 + 32 + 32$
$= 1408$

∴ Total Trainable parameters = $4 \times 1408 = \underline{5,632}$

Question-3

(a) Compute scaling ...., after k = 50 steps.

For k = 10 steps

Scaling Factor = 0.9^{10}
≈ 0.3487

For k = 50 steps

Scaling Factor = 0.9^{50}
≈ 0.0052

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(b) For k = 50, compute ..... for f = 0.99

Ans For f = 0.5

Scaling Factor = $0.5^{50} \approx 0.8882 \times 10^{-15}$

For f = 0.99

Scaling Factor = $0.99^{50} \approx 0.6050$

(c) Explain the result?

Smaller forget gate values causes the gradient to vanish quickly over 50 steps, which makes learning long-term dependencies nicely impossible

LSTM solve this by learning to keep the forget gate value close to 1.0 when memory needs to be preserved, which creates a direct linear highway for the gradient to flow backward through the cell state.

Vanilla RNN lack this path & multiply gradients through repeating nonlinear functions, causing them to vanish completely

Question-4

(a) Total Hourly Records = Years x Day/year x Hour/day

= 3 x 365 x 24

= 26280

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(b) Total Input/Output Window

Total Window Size = 168 + 24 = 192 hrs
(lookback) (prediction)

∴ ⇒ Total Records − Total Window size + 1
= 26280 − 192 + 1
Total Windows = 26,089

(c) With Batch size = 64

Shape Format = (Batch Size, Time Step, Features)

Input Batch Shape = (64, 168, 3)

Output Batch Shape = (64, 24, 1) {If predicting only 1 Feature}

If predicting every feature
= (64, 24, 3)

(d) Training Records = 70% of 26280
= $\frac{70}{100} \times 26280$
= 18,396 records

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Training Windows = 18396 − 192 + 1
= 18,205 windows

Mini Batches per Epoch = $\frac{18205}{64}$ ≈ ~~284 mini batches~~

= 284.45
≈ 285 (including partial batch)

(e) Parameter for 1 Gate :
- Input Weights : 64 × 3 = 192
- Riccurent Weight : 64 × 64 = 4096
- Bias : 64

Total Pergate : 192 + 4096 + 64
= 4352

∵ 4 Gates are there
∴ Total Trainable Parameters = 4 × 4352 = 17,408

Question – 5

(a) Min-Max Normalization = $\frac{x - x_{min}}{x_{max} - x_{min}}$

Normalised Value = $\frac{95-20}{120-20} = \frac{75}{100} = 0.75$

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(b) The o/p probability = 0.83
Threshold = 0.5

∵ O/P > Threshold, so model predicts a Failure (Negative Class)

(c) True Positive (TP) = 30 (correctly flagged failure)
False Positive (FP) = 45-30 = 15 (falsely flagged failure)
False Negative (FN) = 40-30 = 10 (missed actual failure)
True Negative (TN) = 1000-30-15-10 = 945
(correctly flagged normal)

∴ Precision = $\frac{30}{45}$ = 0.667 ≈ 66.7%.

Recall = $\frac{30}{40}$ = 0.750 ≈ 75%.

Accuracy = $\frac{30+945}{1000} = \frac{975}{1000}$ = 0.975

≈ 97.5%.

F1 Score = $2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$

$= 2 \times \frac{0.667 \times 0.750}{(0.667 + 0.750)} = 0.7061$ ≈ 70.61%.

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d) The team should prioritise Recall, missing a real pump failure causes costly downtime / catastrophic damage.

Accuracy is misleading here due to severe class imbalance, only 4% of the total dataset contain actual failures. A broken model predicting "Not Failure" every time still gets a high 96% accuracy.